ModelIt!

Teacher's Guide

Why Do You Look Like That?

The Genetic Lottery That Made You, You

Quick Reference

Grade Level	Grade 8
NGSS Standard	MS-LS3-1, MS-LS3-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: MS-LS3-1, MS-LS3-2

Develop and use a model to describe why structural changes to genes (mutations) located on chromosomes may affect proteins and may result in harmful, beneficial, or neutral effects to the structure and function of the organism. Develop and use a model to describe why asexual reproduction results in offspring with identical genetic information and sexual reproduction results in offspring with genetic variation.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model to describe how genes on chromosomes determine trait inheritance patterns.
Disciplinary Core Idea	**LS3.A Inheritance of Traits / LS3.B Variation of Traits** — Organisms reproduce sexually, producing offspring with genetic variation through random allele combination.
Crosscutting Concept	**Cause and Effect** — Students identify how the CAUSE (specific allele combinations) leads to the EFFECT (observable traits in offspring).

CAST Testing Alignment

MULTIPLE CHOICE:

Two parents are both heterozygous (Bb) for a trait. What is the probability that their offspring will show the recessive phenotype?

CONSTRUCTED RESPONSE:

Explain why two siblings with the same parents can look very different from each other. Use the concepts of alleles, dominant/recessive traits, and random combination in your answer.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every human has approximately 20,000-25,000 genes on 23 pairs of chromosomes. The specific combination of alleles you inherit makes you genetically unique — unless you have an identical twin. Understanding inheritance is key to medicine, agriculture, and evolution.

Chromosomes and Genes

Humans have 23 pairs of chromosomes — one set from each parent. Each chromosome contains hundreds to thousands of genes. Each gene has a specific location (locus) on a chromosome. You have two copies of every gene (one from each parent), and these copies may be the same or different versions (alleles).

Dominant and Recessive

Dominant alleles (represented by capital letters, like B) are expressed when at least one copy is present. Recessive alleles (lowercase, like b) are only expressed when two copies are present (bb). Heterozygous individuals (Bb) carry the recessive allele but show the dominant trait — they are 'carriers.'

Sexual vs. Asexual Reproduction

Sexual reproduction mixes genetic material from two parents, creating unique offspring through random allele combination. Asexual reproduction produces genetically identical offspring (clones). Genetic variation from sexual reproduction is crucial for species survival because it provides the raw material for natural selection.

Beyond Simple Inheritance

Most human traits aren't controlled by a single gene with simple dominant/recessive patterns. Traits like skin color, height, and intelligence are influenced by multiple genes (polygenic inheritance) AND environmental factors. This is why human diversity is so rich and complex.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify parent alleles, offspring genotype, and trait expression as system components.

E — Establish Relationships
Students determine that parent alleles combine to create offspring genotype (positive), and genotype determines phenotype based on dominant/recessive rules.

V — Visualize & Model
Students build the inheritance model showing how parental alleles combine in offspring.

E — Evaluate Outcomes
Students run different parent genotype combinations and observe phenotype ratios.

R — Revise & Extend
Students add mutation or environmental factors to explore more complex inheritance.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Look at your hands. Your eyes. Why do they look the way they do?"*

Watch For

Quick trait survey: Who can roll their tongue? Attached vs. detached earlobes?

Slide 2: Learning Objectives

3 min

Say: *"Today we decode the genetic lottery that made you, you."*

Watch For

Pre-teach allele, dominant, recessive with simple examples.

Slide 3: Big Question

3 min

Say: *"You got DNA from both parents. So why aren't you a 50-50 clone?"*

Watch For

Show sibling photos that look very different. Discuss.

Slide 4: LEVER Framework

2 min

Say: *"We'll model how genes are shuffled to create unique individuals."*

Watch For

Preview the LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What goes INTO making your traits? What comes OUT?"*

Watch For

Sort components. Emphasize: one allele from each parent.

Slide 6: Activity 2: Connections

8 min

Say: *"Dominant alleles are like volume — one copy is enough to be 'heard.'"*

Watch For

Connect Punnett square logic to the model relationships.

Slide 7: Activity 3: Simulation

10 min

Say: *"What happens when two carriers have children? Let's see the ratios."*

Watch For

Students predict, then run. Discover the 3:1 ratio.

Slide 8: Discoveries

5 min

Say: *"Every time you're conceived, it's a genetic lottery. The odds are never exactly the same."*

Watch For

Discuss: Why is variation important for species?

Slide 9: STEM Challenge

12 min

Say: *"A rare animal cub was found. CSI: Genetics — figure out who the parents are!"*

Watch For

Teams use trait evidence and Punnett squares to solve the mystery.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Parent 1 Alleles, Parent 2 Alleles
- **INTERNAL:** Offspring Genotype, Trait Expression (Phenotype)

Relationships

- **Parent 1 Alleles to Offspring Genotype:** POSITIVE (+) — Parent 1's alleles directly contribute to half of the offspring's genetic makeup.
- **Parent 2 Alleles to Offspring Genotype:** POSITIVE (+) — Parent 2's alleles contribute the other half of the offspring's genetic makeup.
- **Offspring Genotype to Trait Expression:** POSITIVE (+) — The specific allele combination determines which phenotype is expressed based on dominant/recessive rules.

Key Discoveries

- You inherit ONE allele from each parent — your unique combination is why you're genetically unique
- Dominant alleles mask recessive ones, but recessive alleles are still passed on (carriers!)
- The classic Bb × Bb cross produces a 3:1 phenotype ratio (75% dominant, 25% recessive)
- Sexual reproduction creates variation through random allele combinations — that's evolution fuel

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Traits blend like paint"	Traits don't blend — alleles maintain their identity. A Bb individual doesn't have 'medium' expression. The dominant allele is expressed and the recessive is masked but preserved.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"You look exactly 50% like each parent"	You inherit 50% of your DNA from each parent, but WHICH 50% is random. Siblings (except identical twins) get different random halves, which is why they can look very different from each other.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Dominant means better or more common"	Dominant only means the allele is expressed when one copy is present. It doesn't mean it's 'better' or more frequent. Polydactyly (extra fingers) is dominant but rare.